# Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

**Weekly Progress Report — Week 7**

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## 1 Introduction

Week 7 focused on resolving physical conveyor distortions: **fabric web angular skew and conveyor misalignment**. In continuous industrial loom operations, fabric rolls move along conveyor rollers under high-speed line-scan inspection. Conveyor vibration, roller tension shifts, and tracking drift introduce a slight angular tilt between the fabric yarn axes and the camera sensor array.

Because directional texture operators and vertical column profiling assume that threads are strictly aligned with image rows and columns, angular tilt causes filter footprints to cut obliquely across adjacent yarn channels. This dilutes defect contrast and produces false boundary responses. To eliminate this issue, we developed an automated, unsupervised web de-skewing module based on the **Radon Transform projection variance**.

## 2 Radon Projection Variance Principle

The Radon transform integrates pixel intensities along straight lines at specified angles across the image:

### 2.1 Projection Contrast Maximization

When the projection angle matches the true orientation of warp or weft threads, integration lines run directly along yarn ridges and inter-thread gaps. This produces a projection profile with sharp peaks (dense threads) and deep troughs (spaces between threads), resulting in maximum sample variance.

Conversely, if the projection angle is misaligned, each line cuts across multiple threads, averaging bright fibers and dark valleys together into a flat, low-variance profile. By searching for the angle that maximizes this projection variance, the true web orientation is determined reliably without requiring edge detectors or manual landmark points.

## 3 Coarse-to-Fine Search Optimization

To make the alignment practical for real-time inspection, we engineered a two-stage search strategy:

- **Coarse Angular Scan:** The system initially evaluates projection variance over a broad angular range at one-degree increments on a downsampled central crop of the fabric. This rapidly isolates the primary orientation peak.
- **Fine Angular Refinement:** The search window is then narrowed to a tight bracket around the coarse peak. A fine sweep evaluates projection variance at fractional-degree steps to extract sub-degree tilt estimates.
- **Active Fabric Boundary Masking:** Line-scan images often contain dark conveyor margins or selvedge borders. Horizontal edge profiling isolates the valid fabric region so that external borders do not bias the angle estimation.

## 4 Affine Rectification and Boundary Handling

Once the optimal tilt angle is identified, the image coordinate frame is restored:

- **Bicubic Affine Transformation:** The image is rotated around its geometric center using bicubic interpolation, preserving delicate yarn textures and preventing pixel jaggedness.

- **Peripheral Artifact Blanking:** Rotating an image naturally creates empty wedge regions along the borders. The module dynamically generates a valid boundary mask, ensuring downstream defect detection ignores outer interpolation margins.

## 5 Work Completed This Week

- Formulated the Radon projection variance concept for unsupervised fabric orientation detection.
- Implemented the modular alignment engine in `src/preprocessing/alignment.py`.
- Developed the coarse-to-fine search framework to achieve high precision with low computational overhead.
- Built active fabric border detection and bicubic affine rotation with border masking.
- Created a standalone test script (`code/scripts/run_phase1_alignment.py`) applying synthetic rotations to verify angle recovery across different weaves.

## 6 Plan for Next Week

With geometric web alignment established, next week will focus on periodic background weave suppression:

- Investigate 2D Fourier frequency-domain analysis to model repetitive yarn patterns.
- Develop automated frequency peak detection to isolate dominant weave harmonics.
- Implement 2D Butterworth cross-notch rejection filters to suppress background patterns before spatial morphology.

*End of Week 7 Report*